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Innovative Workflows for Brown Field Revitalization - Case Study from Libya

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Abstract

Over the years, hundreds of oil wells in the largest oil field in Libya have been depleted, leaving considerable oil volumes behind due to strong aquifer support and improper water encroachment management. A comprehensive intervention program was developed to evaluate and produce this trapped oil. Conventional electric submersible pump (ESP) designs did not consider changes in hydrocarbon saturation. Therefore, tailored Y-tool systems were used to achieve inflow profile passes and avoid undesired nonproductive rig time.

The intervention workflow involved a thorough review of the workover histories of nearly 100 wells, revealing prevalent well integrity issues and solids production. Many wells experienced a gradual decline in oil production rates, leading to frequent ESP failures. To tackle these challenges, a comprehensive workover program was devised. This program involved retrieving failed pumps, assessing wellbore integrity, installing ESP bypass systems, and conducting casedhole logging for saturation monitoring and flow profiling. Multirate and pressure transient tests were performed to accurately assess reservoir performance, informing subsequent remedial actions. This integrated approach enabled dynamic reservoir evaluation, facilitating the identification of bypassed and remaining hydrocarbons, damaged areas, watered-out zones, and well integrity issues.

The comprehensive evaluation provided insights crucial for the accurate design of ESP systems tailored to each well's potential. This level of precision, previously unattainable due to the absence of measurements of the dynamic reservoir conditions, underscores the significance of the integrated methodology in optimizing well performance and mitigating ESP failures. Through the implementation of this approach, over 100 previously shut-in and underperforming wells were effectively restored. These efforts resulted in a remarkable surge in oil production, with a gain of nearly 63,000 barrels of oil per day (BOPD), coupled with a substantial reduction in water production exceeding 90,000 barrels of water per day. These outcomes validated the effectiveness of the project and led the operator to embark on a subsequent phase. This next step seeks to capitalize on the success of the initial endeavor and embrace further opportunities for progress and advancement.

Introduction

The Sarir oil field, located in the southeastern part of the Eastern Embayment of the Sirt Basin, is Libya's largest oil field in terms of conventionally recoverable reserves and a major contributor to Libya's oil production. As shown in Fig. 1, the Sarir Main field consists of several tilted fault blocks forming a large, faulted anticline situated between the Sarir Trough and the South Sarir Ridge. The reservoir is primarily located in the Lower Cretaceous Sarir Sandstone Formation, which is progressively truncated beneath the mid-Cretaceous unconformity toward the northwest margin. Production from Sarir Main began in December 1966, and by 2013, cumulative production exceeded 3,000 MMB, resulting in a recovery factor greater than 36%. This high recovery factor is attributed to the exceptional qualities of the Sarir Sandstone, which is believed to support conventional recovery up to 46%. The primary artificial lift method utilized in the field is electric ESPs, which have been essential in managing the high water cut that averages over 70%. The field does not have a gas cap, and the gas/oil ratios vary between 60 and 225 SCF/STB, presenting challenges for pressure maintenance. Water injection has been employed to maintain reservoir pressure and production rates.

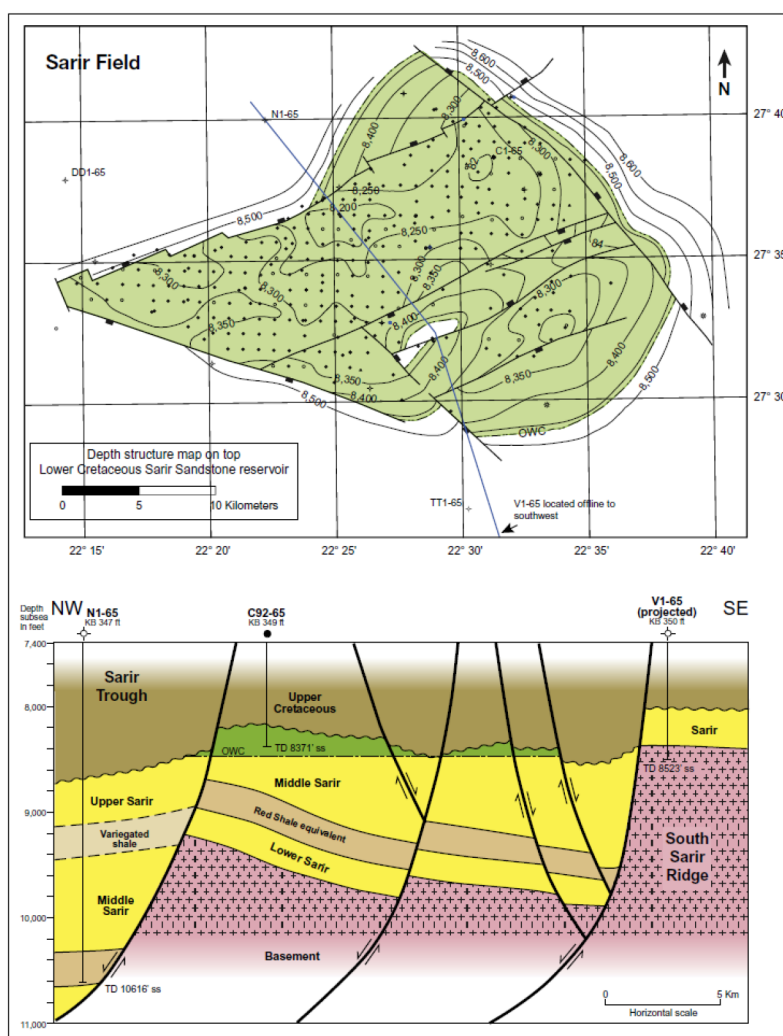


Figure 1—Depth structure map on top Lower Cretaceous Sarir Sandstone reservoir of Sarir Main Field (also known as Sarir C) with northwest to southeast cross section.

The field has faced significant operational challenges, including severe sanding problems that affect ESP performance and well productivity. Additionally, the presence of a tar mat, with thickness ranging from 10

to 70 ft, and the high wax content of the oil (19%) pose additional complications for future development plans. Despite these challenges, the field remains a critical asset for Libya's oil industry, with ongoing efforts to enhance recovery and maintain production levels.

The Messla Field, ranking seventh among Libya's giant oil fields in terms of reserves, is located 30 km north of the Sarir Main Field within the same geological setting of the Sirt Basin. The field is a classic stratigraphic trap with oil trapped in the Sarir Sandstone, which wedges out onto the eastern flank of the Messla High, as shown in Fig. 2. The Messla Field has an original oil in place of approximately 3,000 MMB, and by 2005, production had reached 1,105 MMB, yielding a recovery factor of 37%. The reservoir's excellent fluvial characteristics are expected to allow conventional recovery as high as 53% in the northern part of the field and 47% in the southern part (Ahlbrandt 2001).

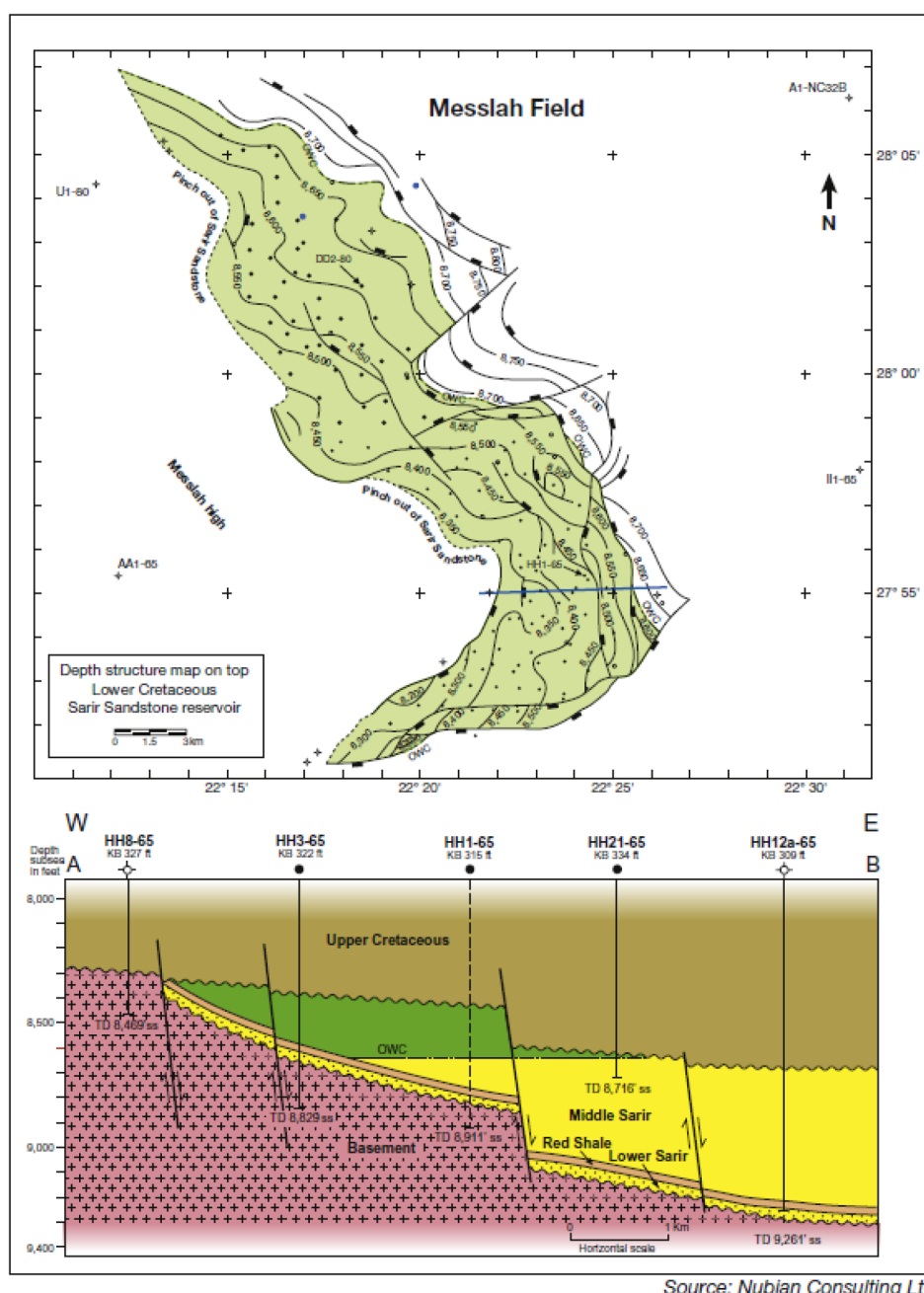


Figure 2—Depth structure map on top Lower Cretaceous Sarir Sandstone reservoir with west to east cross section. Cross section is through reservoir interval with vertical exaggeration $\times 10$. Original oil/water contact.

Production at Messla peaked at 152,000 BOPD in the late 1980s, and by 2010, the field was producing around 120,500 BOPD. The primary artificial lift system employed in Messla is ESPs, which are crucial in managing the field's high water cut, which averages above 70%. The wax content of the oil in Messla is lower than in Sarir, at approximately 8%, and the field's sulfur content is reported to be 0.99%.

The Messla Field has encountered significant operational challenges, particularly in maintaining production levels amidst high water cut and waxing issues. The field's production was severely impacted by the 2011 conflict, which damaged the surface facilities. Although production resumed by September 2011, the field has since experienced fluctuating output. Future development plans will need to address these challenges to optimize recovery and sustain production.

Objectives

The National Oil Corporation (NOC) has set an ambitious goal to increase Libya's overall oil production to 2 million BOPD by 2027. The Sarir and Messla Fields, which boast the highest original recoverable reserves in Libya are central to achieving this objective. Enhancing the recovery factor in these fields by even a few percentage points could yield a significant and cost-effective increase in oil production.

Given that the Messla and Sarir Fields are 100% owned by the NOC, any production increase directly benefits the government's revenues. However, past workover operations aimed at boosting production prior to 2017 were largely unsuccessful, primarily due to the lack of dynamic surveillance. These operations often failed to address critical issues such as damaged perforated intervals, poor detection of water entry zones, and other uncertain reservoir parameters.

To address these challenges and maximize oil recovery, the project focuses on implementing a comprehensive intervention program. This program involves the use of tailored Y-tool systems in electric ESP operations, designed to account for changes in hydrocarbon saturation and improve inflow profiling. The ultimate goal is to produce previously trapped oil while minimizing nonproductive rig time, thereby contributing significantly to Libya's production targets.

Well Selection

The wells selected for the workover program underwent a comprehensive screening process. The selection criteria focused on shut-in wells and underperforming wells with the aim of increasing oil production. The selection criteria were both technical and operational. Key considerations included the wells' proximity to geological features such as faults and the edges of aquifers, which could influence reservoir behavior and fluid dynamics. Additionally, the performance of each well was thoroughly compared with that of offset wells to identify candidates with the highest potential for enhanced recovery.

A comprehensive review of each well's history was conducted, encompassing completion and workover records, the types of casing used, and the intervals that had been perforated. Special attention was given to the methods employed for isolating certain perforations, such as the use of bridge plugs or cement squeezes, to ensure the integrity of the wellbore and the effectiveness of previous interventions.

Furthermore, an in-depth petrophysical evaluation was carried out, alongside well-to-well analysis, to assess the remaining hydrocarbon reserves and monitor the movement of the oil-water contact. This analysis was crucial for understanding the current state of the reservoir and planning the most effective workover strategies to maximize recovery.

Operational Challenges and Solutions

To conduct dynamic surveillance in the Sarir and Messla Fields, where most wells are completed with 7-in. casing and ESPs without a bypass system, operators face the challenge of performing surveillance without rig intervention, which typically results in significant nonproductive time and lost production. The successful deployment of the Y-tool in these fields has provided a solution to this challenge, enabling

surveillance and intervention activities without the need for complete workover operations. However, the implementation of tailored Y-tool systems in ESP operations introduced several technical challenges, particularly in addressing changes in hydrocarbon saturation and navigating well geometry complexities. Furthermore, minimizing damage induced by workover operations both before and after surveillance and intervention remains critical for optimizing well performance and maintaining the integrity of the production system.

Example 1: Unsuccessful Retrieval of 2 3/4-in. Blanking Plug

In this case, an advanced ESP and 9 5/8-in. Y-tool assembly, including dual pumps and a 540 series motor, was deployed. Following the completion of run in hole (RIH) operations, a blanking plug was set within the tele-swivel nipple using a slickline (SLK) running tool to facilitate well clean-up procedures. However, when the time came to retrieve the blanking plug, multiple attempts were unsuccessful, necessitating the retrieval of the plug by pulling out of the hole (POOH) and dismantling the Y-tool assembly. Post-retrieval inspection revealed that the plug was obstructed by sediment and debris accumulation, as shown in Fig. 3; this had completely sealed the equalization ports, thereby inhibiting pressure equalization and complicating the retrieval process. The primary cause was identified as sand accumulation within the wellbore, which impeded the proper functioning of the equalizing sleeve. To prevent recurrence, it was recommended that the production tubing be thoroughly cleaned and drifted prior to assembly and that the wellbore be adequately scraped and cleaned before initiating operations. Furthermore, avoiding the use of check valves in wells intended for production logging or reservoir saturation evaluation runs and instead utilizing standing valves, which offer a more reliable mechanism, is advised.



Figure 3—Accumulation of sediment and debris accumulation in the retrieved blanking plug.

Example 2: Wireline Logging Tools Stuck Due to Bent 2 1/8-in. Bypass Tubing

In this case, a comprehensive ESP and 7-in. Y-tool assembly was installed, featuring two slim pumps, a slim rotary separator, a protector, a single 375 series motor, and five bypass tubes. The well's total depth was recorded at 9,150 ft, with a 7-in. casing (#26) and a pump setting depth of 7,582 ft. During the production logging tool (PLT) run, the wireline tool was unable to pass the wireline reentry guide (WEG) and became stuck at a depth of 7,606 ft. Subsequent attempts to clear the obstruction by running a blanking plug for three hours and deploying a 1 3/4-in. gauge cutter to 7,700 ft were unsuccessful, with the PLT again becoming stuck at 7,606 ft. After further attempts, the decision was made to POOH, revealing that bypass tubing joints 3 and 4 were visibly bent. The root causes were identified as the installation of a finned WEG with an outer diameter of 3 4/5 in. in a 7-in. casing (#26), incorrect positioning of the long fin during assembly, potential misalignment of the rig, and inadequate casing cleaning. Best practices to mitigate these issues

include ensuring precise rig alignment, drifting the 7-in. casing with an external drift assembly, avoiding the use of finned WEGs, and instead using an open-end 2 1/8-in. bypass joint with a 45° angle cut.

Example 3: Y-Tool Assembly Stuck During RIH

A significant challenge was encountered during the RIH of a 7-in. Y-tool and slim ESP assembly with 2 3/8-in. bypass tubing inside a 7-in. casing (#26). During the operation, the entire assembly became stuck at a certain depth, halting progress. Upon POOH and investigation, it was determined that the well contained mixed casing weights—specifically, sections of 7-in. casing (#29) were present alongside the #26 casing. This discrepancy was not accounted for due to inaccurate well geometry data provided by the operator and insufficient cleaning of the casing prior to RIH. The primary causes included a lack of accurate well geometry information, inadequate casing cleaning, and the use of an undersized scraper that failed to detect the heavier casing sections. To address these issues, it is essential to ensure the 7-in. casing is thoroughly drifted and cleaned before assembly, verify well geometry data with the operator, and always use a scraper compatible with the casing diameter. Additionally, the 2 1/8-in. bypass tubing should be employed to minimize risks associated with wellbore data inaccuracies, and specialized equipment configuration and RIH procedures should be adhered to, as outlined in [Hammad et al. \(2023\)](#).

Improved Workover Program

The improved workover program began with the deployment of a workover rig to replace the existing failed ESPs. Following this, a comprehensive integrity test was conducted to ensure the absence of any integrity issues that could affect subsequent operations. In the event of integrity issues, full remedial work was conducted. Once integrity was confirmed, an ESP test pump, equipped with a bypass system compatible with the completion size, was run into the well. A cleanup period was then performed to restore the near-wellbore saturation profile.

Subsequently, a SLK operation was carried out to retrieve the blanking plug. This was followed by the deployment of a combination of production logging and reservoir saturation tools to evaluate the remaining hydrocarbons and flow profile. This evaluation aimed to identify the source of water production, whether it originated from the formation, a leak behind the casing, or the plug back.

A multirate test was then conducted to construct an inflow performance relationship curve for all layers. This curve would be used to optimize the new ESP pump following water shutoff procedures. Additionally, a buildup test was occasionally conducted to assess the overall reservoir properties and skin. More details and example cases are provided in the next section.

Case Histories and Lessons Learned

Since the initiation of the new improved workover program in 2019, more than 100 wells have been successfully executed. This program has provided numerous valuable lessons and has significantly increased oil production for the Arabian Gulf Oil Company. Below are some notable cases to illustrate these achievements:

Case 1: The Importance of Downhole Surveillance for Production Enhancement

The newly developed workover procedures were designed to facilitate the use of production logging and reservoir saturation tools under dynamic conditions. This enhanced program enabled the operator to precisely identify water sources, detect flow behind the casing, locate bypassed zones, and recognize underperforming layers, among other advantages. Production logging and saturation logs for one of the wells are displayed below, clearly indicating depleted zones and water entry points, as presented in [Fig. 4](#). Consequently, the decision was made to isolate the bottom layer and stimulate the upper layers to enhance oil production.

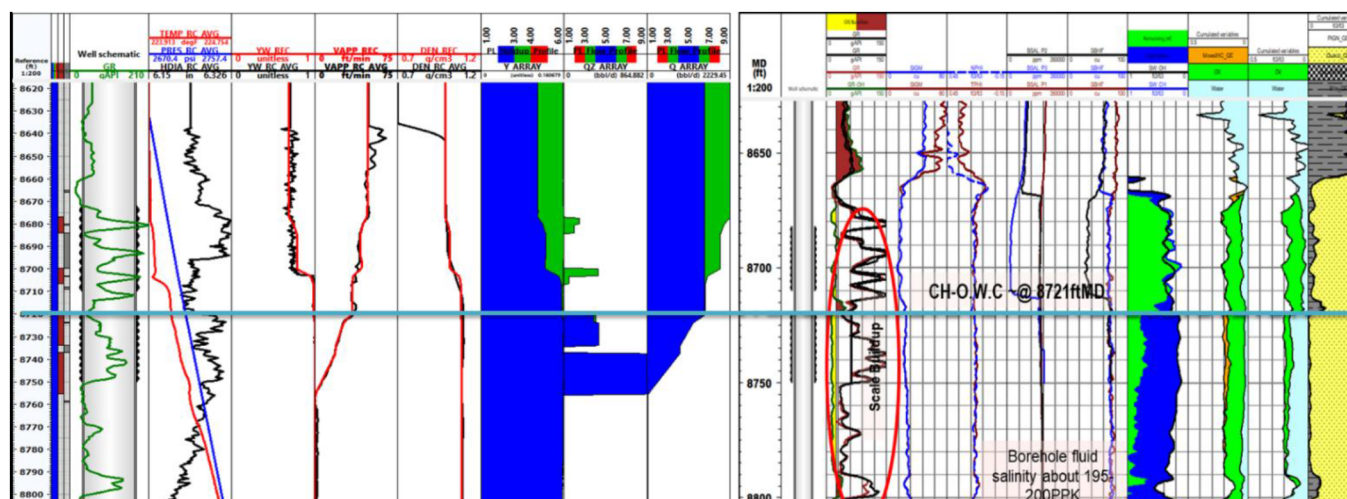


Figure 4—An example of a successful water shut-off operation was performed during the project

Case 2: The Importance of Combining Static and Dynamic Logs for Downhole Surveillance

Static and dynamic surveillance complement each other and are equally important. Dynamic data from production logging provides insights into water and oil entry points, flow behind casing, and active versus inactive zones. However, when inactive zones or flow behind casing are identified, production logging alone cannot determine whether these layers still have hydrocarbon potential.

In contrast, saturation logs offer insights into the hydrocarbon in place but cannot distinguish between movable and non-movable oil. Therefore, integrating both logs is crucial to compensate for each tool's shortcomings, especially for water shutoff procedures. In the example below, production logging indicates that all perforations produce water, with a small hydrocarbon entry point (~2%) from the top of the bottom perforation. However, the saturation log suggests that the top of the bottom perforation and the upper two perforations still have significant hydrocarbon potential.

Upon further analysis, including comparing shut-in logs, flowing logs, and cement logs, it was discovered that flow behind casing was affecting the oil entry points. The decision was made to repair the cement across the zone of interest and isolate the bottom of the bottom perforation to enhance oil production.

Case 3: Casing Leak Detection and Isolation

Due to expected large water production from Well HH-07, a comprehensive well cross-section analysis was conducted to evaluate the stratigraphy and various petrophysical properties of the well. This analysis involved a detailed comparison with nearby wells to identify similarities and differences in geological formations and reservoir characteristics (refer to Fig. 6). The study aimed to enhance the understanding of the subsurface conditions, which is crucial for optimizing well performance.

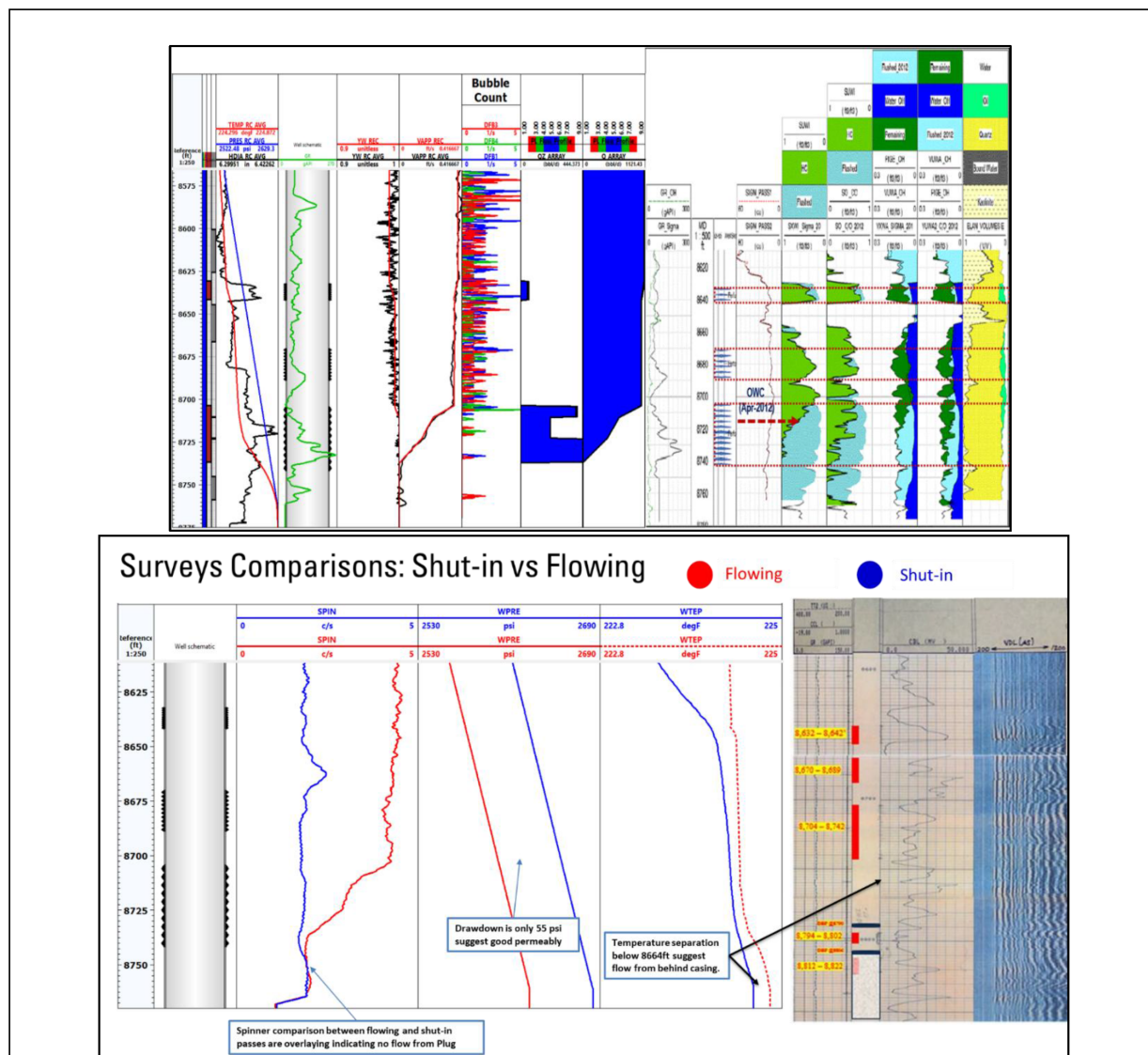


Figure 5—An example of production logging and reservoir saturation integration to determine a flow behind the casing.

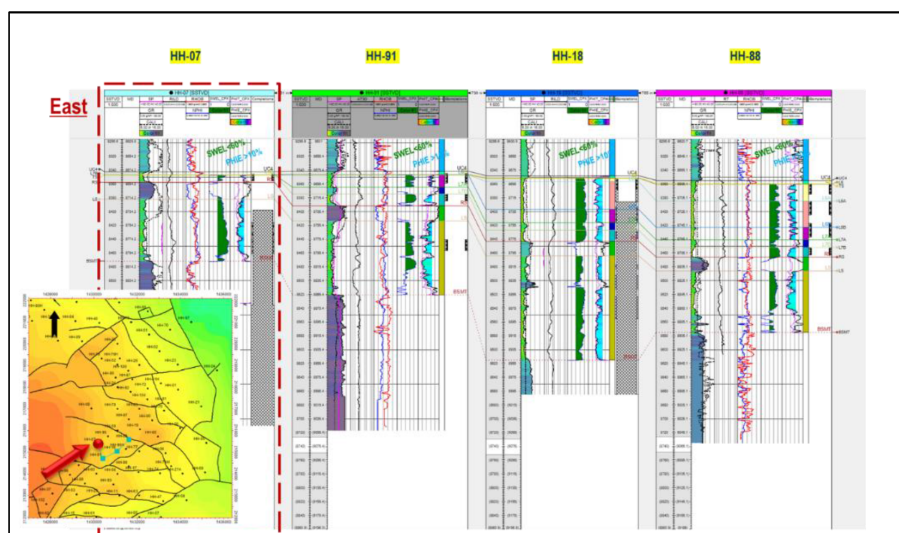


Figure 6—Comprehensive well cross section analysis.

HH-07 Workover History. Well HH-07, completed in January 1975, initially produced oil naturally until 1994, after which it faced multiple operational challenges, primarily due to casing leaks and increasing water production. Several workovers were conducted, including cleaning the well, reperforating zones, and installing ESPs. Despite these efforts, the well experienced a rapid increase in water cut, as illustrated in Fig. 6, which led to suspensions and multiple unsuccessful casing repairs.

Casing evaluation using an ultrasonic tool (explained in Fig. 7) identified deformation that could not be repaired by cement, and a leak at 6,965 ft was addressed unsuccessfully. Subsequent attempts to isolate the leak using traditional methods were unsuccessful due to well conditions.

To overcome these challenges, a new approach was adopted, utilizing an ESP Y-tool and ESP packer. The casing leak was isolated effectively, and a slim ESP was installed, allowing continued operation without further compromising production. Reservoir saturation tool logs showed significant oil remained in the Lower Sarir formation, and the subsequent intervention confirmed that the high-water cut was primarily due to shallow zone leaks rather than reservoir coning. After isolating the leaking zones and keeping the productive perforations open, the well's performance improved dramatically, with oil production rising to 1,000 BOPD and water cut reduced to just 2%. This intervention restored productivity and minimized water production, demonstrating the effectiveness of leak isolation and optimized ESP technology.

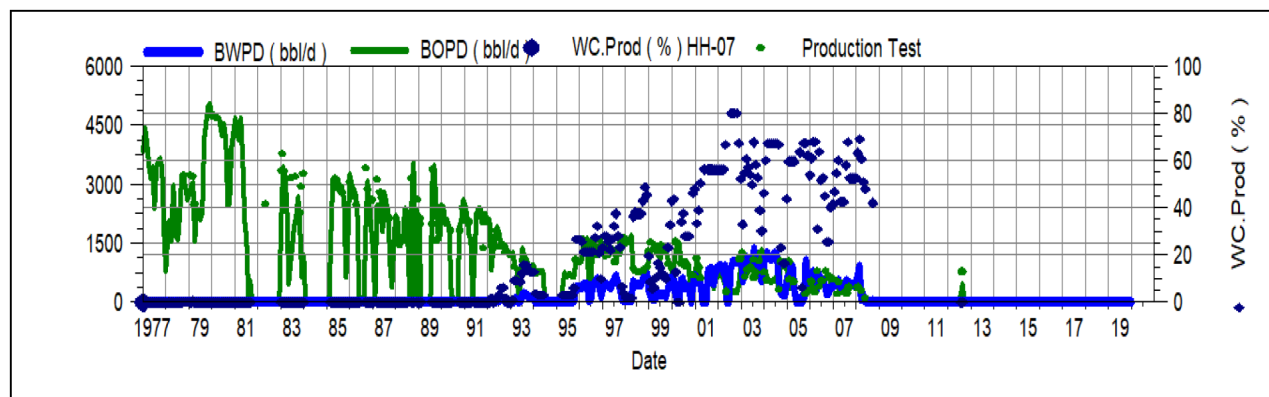


Figure 7—HH-07 production history.

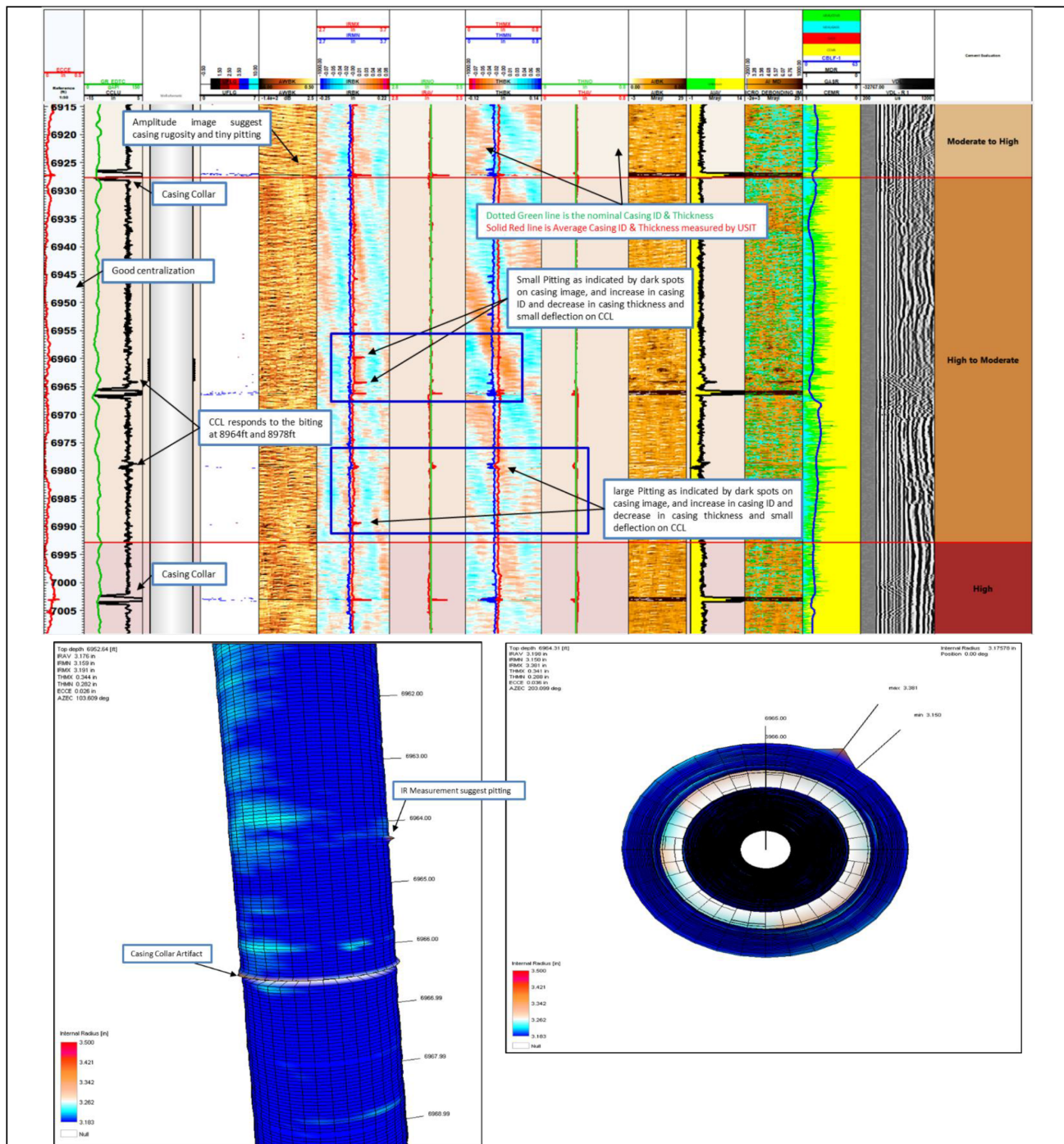


Figure 8—Ultrasonic tool casing evaluation revealing deformation at 6965 feet.

Case 4: Multirate Tests and Buildup

Test Planning and Procedures. The following procedure outlines the steps for conducting a multirate test to evaluate well performance using an ESP. Below is an example of a test workflow that was conducted on the candidate well L-27. The productivity index (PI) of this well was uncertain, which is a critical parameter for the design of a permanent ESP system:

- Pressures are recorded during a buildup phase and during successively increasing rate steps as the ESP speed is ramped up.

- Both rates and pressures should be recorded during the flow periods as well.
- After the rate and pressure have stabilized, the rate is then increased. This process is repeated as desired (usually four rates for a four-point test), and then the well is shut-in again for a final buildup.
- Once the data has been gathered, the bottomhole pressures are plotted on an absolute open flow (AOF) plot to determine the deliverability of the well and the AOF.
- The flow rate is to be tested using a multiphase flow meter (MPFM).

A slim ESP pump, as shown in Fig. 9, was selected to conduct the test. It is noteworthy that the string is equipped with a Y-tool, which facilitates wireline logging. Additionally, the ESP is outfitted with downhole ESP gauges to measure both pressures and temperatures at the pump intake and discharge points. These measurements, along with the production flow rate data obtained from the MPFM, will be utilized to comprehensively analyze the test results.

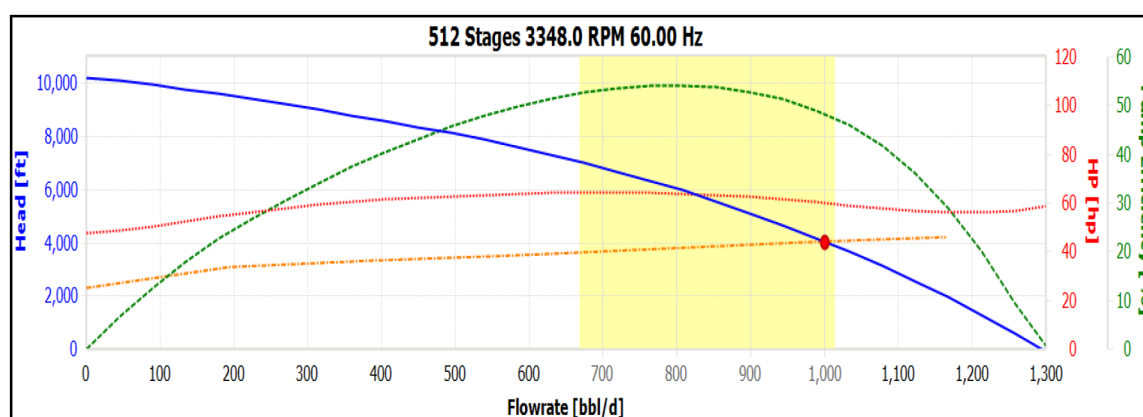


Figure 9—Actual pump performance chart of a slim test pump.

Test Raw Data. The plot below illustrates the production data and ESP downhole gauge parameters recorded during the well test. Initially, the well underwent a cleanup period to ensure the removal of any debris or unwanted fluids. Following this phase, the flow rate was measured continuously until stable readings were obtained for a minimum duration of six hours. Subsequently, a multirate flow test was conducted by varying the ESP frequency to modulate the flow rate. During this test, the corresponding intake pressure was meticulously recorded. Upon completion of the flow test, the ESP was shut down, and the well valves were closed at the surface. Additionally, the variable speed drive remained operational to monitor and record the pressure buildup over time. The multirate flow test data was analyzed using straightline methods, as shown in Fig. 7, resulting in an estimated PI of 2.37 STB/D/PSI.

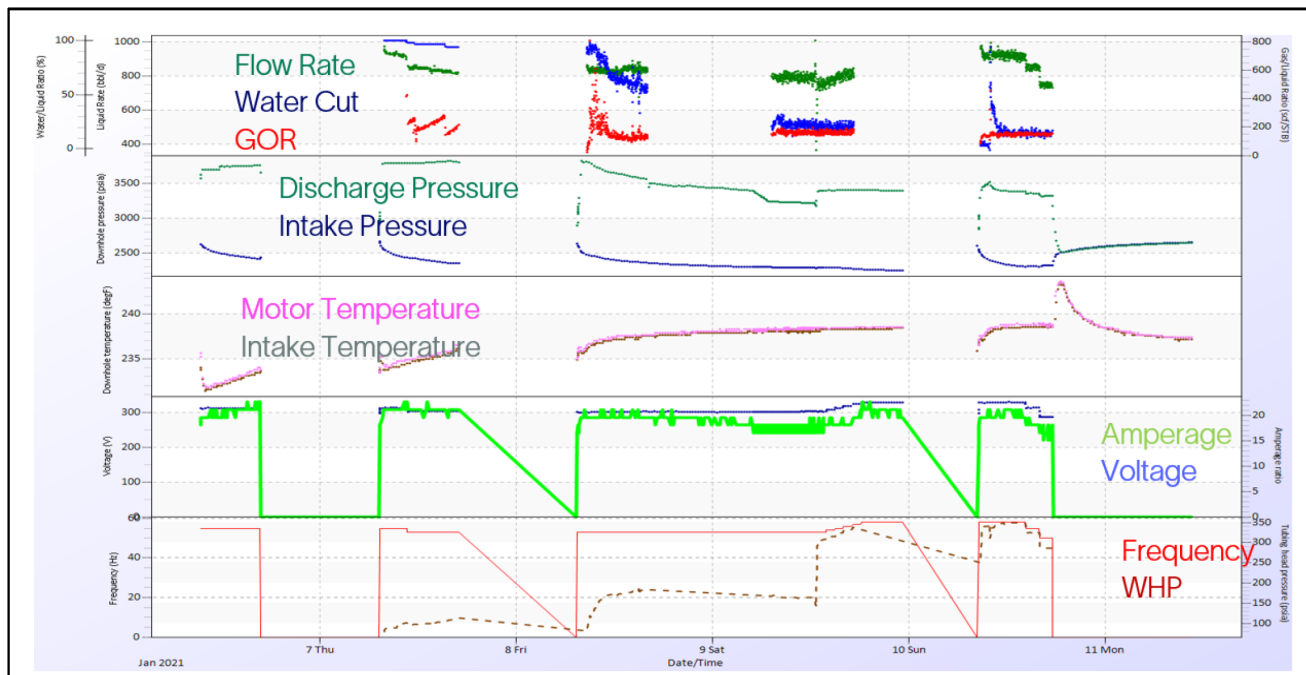


Figure 10—Production and ESP downhole gauge data which were recorded during the flowing period.

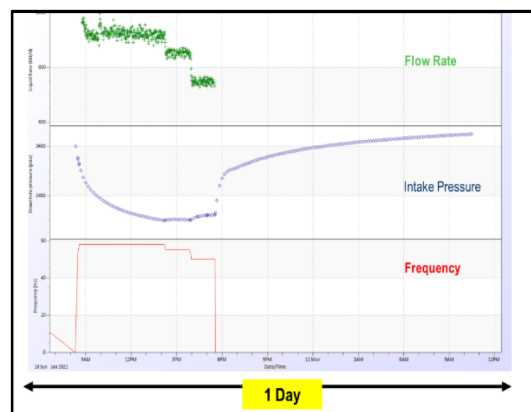


Figure 11—Buildup data that was recorded during the shut-in period.

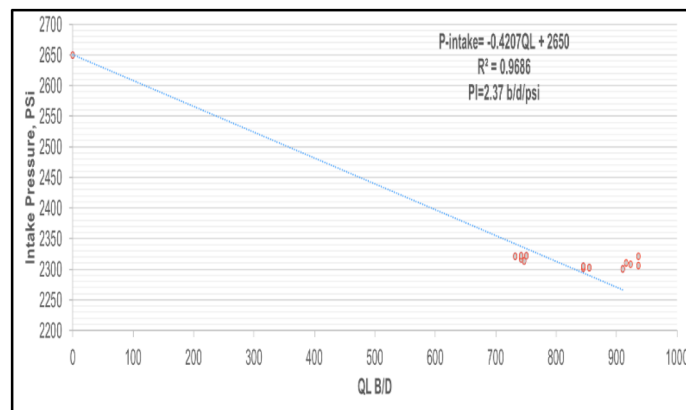


Figure 12—Productivity index estimation using flowing and shut-in test points.

ESP Diagnostics. The diagnostic analysis of an ESP was conducted using data from the ESP downhole gauge, which provided crucial insights into the pump's performance. Additionally, the flow rate and water cut were measured using an MPFM. The combined data indicated that the ESP was operating within its recommended range, ensuring optimal performance and efficiency. The analysis revealed that the ESP was functioning normally, with no signs of abnormal behavior or potential issues. The PI, a key indicator of the pump's efficiency, was calculated to be approximately 2 barrels per day per psi (b/d/psi). This value suggests that the ESP is effectively converting pressure into fluid flow, maintaining a stable and productive operation.

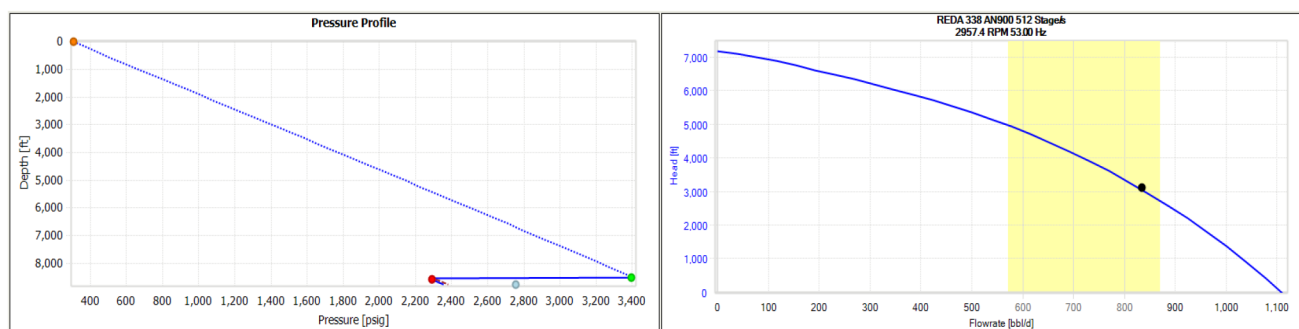


Figure 13—Example of ESP diagnostic charts after matching the design data with the actual data from the test

Pressure Transient Analysis. Pressure transient analysis (PTA), as shown below, can also be performed using well test shut-in pressure data and production history. Radial flow was detected, and the permeability was calculated to be 52 md. The estimated skin factor indicated well stimulation as a result of new perforations. The shut-in pressure at gauge depth (8,678 ft) was 2,756 psi, and the absolute open flow potential was estimated at 5,510 stock tank barrels per day (STB/D).

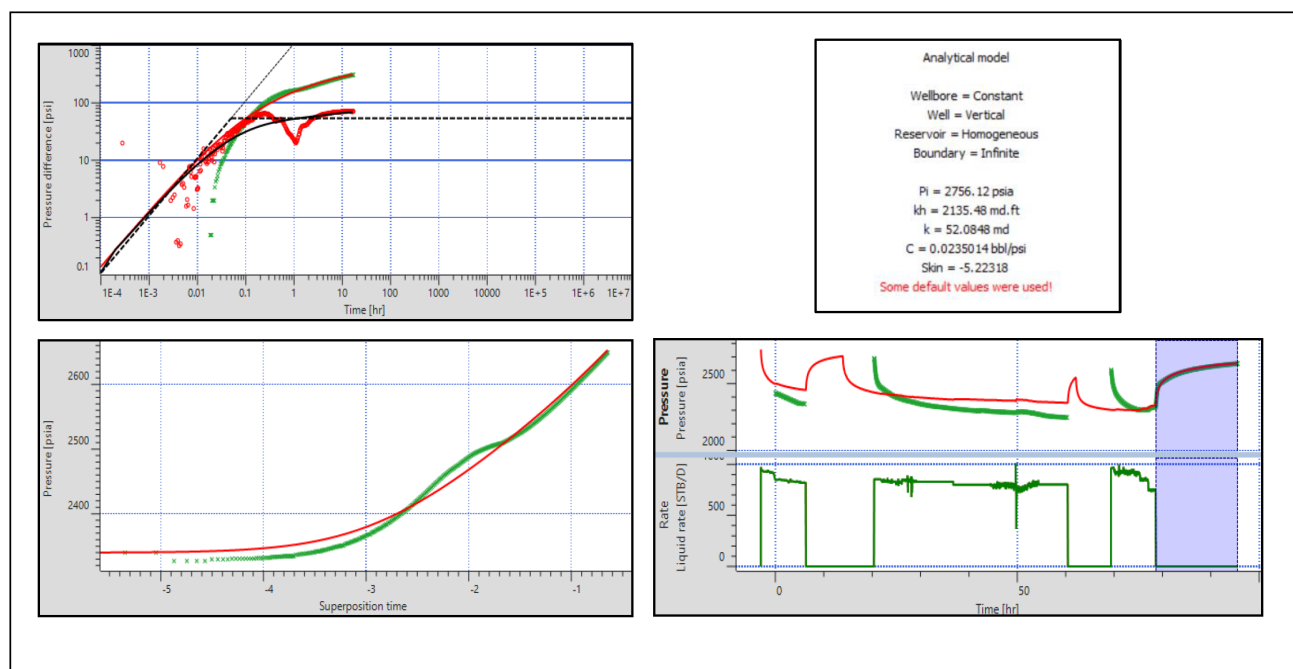


Figure 14—Pressure transient analysis performed using the ESP gauge parameters and surface flow pressures and rates

Conclusions

In conclusion, the enhanced workover program, which employed advanced ESP technology and innovative dynamic surveillance strategies, has proven highly effective in optimizing well performance and boosting oil production. The approach involved the deployment of Y-tool systems to facilitate interventions without the need for complete workover operations, significantly reducing nonproductive time and preserving production continuity. This improvement was particularly crucial in wells completed with 7-in. casings without a bypass system, enabling efficient monitoring and subsequent actions without the associated downtime. The integration of production logging and reservoir saturation tools provided valuable insights into well performance, including identifying water entry points, flow behind casing, and underperforming zones. This data-driven approach enabled targeted isolation of problematic zones, such as shallow leaks, while keeping productive intervals open, thereby improving oil output and minimizing water production.

Furthermore, rigorous pre-workover screening criteria—including an in-depth review of historical well data, petrophysical evaluations, and comparisons with offset wells—ensured the most suitable candidates were selected for intervention. The case histories presented highlighted key lessons learned, such as the criticality of proper casing cleaning, accurate well geometry information, and the use of suitable equipment configurations to avoid operational setbacks. Detailed analysis, such as PTA and multirate testing, played a significant role in determining optimal inflow performance and understanding reservoir properties, which directly informed ESP design and operational settings. Ultimately, the successful implementation of this improved workover program across over 100 wells since 2019 has led to a substantial increase in oil production for the Arabian Gulf Oil Company. This experience underscores the importance of employing an integrated approach—combining robust diagnostics, precision in execution, and lessons learned from past interventions—in enhancing reservoir recovery and extending the operational lifespan of artificial lift systems in challenging environments.

Nomenclature

VSD	Variable Speed Drive
ESP	Electric Submersible Pump
BHP	Bottomhole Pressure
AOF	Absolute Open Flow
MPFM	Multiphase Flow Meter
IPR	Performance Relationship
WSO	Water Shutoff
GOR	Gas/Oil Ratio

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